LANGUAGE DIVERSITY: EVALUATING LANGUAGE USAGE AND AI PERFORMANCE ON AFRICAN LANGUAGES IN DIGITAL SPACES

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ABSTRACT

This study examines the digital representation of African languages and the challenges this presents for current language detection tools. We evaluate their performance on Yoruba, Kinyarwanda, and Amharic. While these languages are spoken by millions, their online usage on conversational platforms is often sparse, heavily influenced by English, and not representative of the authentic, monolingual conversations prevalent among native speakers. This lack of readily available authentic data online creates a challenge of scarcity of conversational data for training language models. To investigate this, data was collected from subreddits and local news sources for each language. The analysis showed a stark contrast between the two sources. Reddit data was minimal and characterized by heavy code-switching. Conversely, local news media offered a robust source of clean, monolingual language data, which also prompted more user engagement in the local language on the news publishers' social media pages. Language detection models, including the specialized AfroLID and a general LLM, performed with near-perfect accuracy on the clean news data but struggled with the code-switched Reddit posts. The study concludes that professionally curated news content is a more reliable and effective source for training context-rich AI models for African languages than data from conversational platforms. It also highlights the need for future models that can process clean and code-switched text to improve the detection accuracy for African languages.

1 Introduction

The African continent is characterized by immense linguistic diversity, with over 2,000 languages spoken by a population of more than 1.5 billion people [1]. This rich tapestry of languages facilitates cultural exchange and communication across a vast array of communities. However, the representation of African languages in the digital sphere has not kept pace with advancements in technology. While language technologies, particularly Large Language Models (LLMs), have demonstrated remarkable capabilities for widely-spoken global languages, their performance on African languages remains a significant challenge [2]. To investigate this issue, we evaluate the digital presence of African languages on conversational platforms and assess the performance of a modern AI tool in identifying them. Our investigation focuses on three prominent languages:

- **Yoruba:** A West African language with over 50 million speakers primarily in Nigeria, Benin, and Togo. Its influence also extends to diaspora communities in the Americas, including Brazil, Cuba, and Trinidad.

- **Kinyarwanda:** A Bantu language spoken by approximately 12 million people, primarily in Rwanda, where it is one of the four official languages alongside English, French, and Swahili.
- **Amharic:** The official working language of Ethiopia's federal government and the second-most spoken Semitic language after Arabic, with over 57 million speakers. It belongs to the Afro-Asiatic language family.

2 Problem Statement

Despite having millions of speakers, the digital footprint of these languages on online platforms is notably limited. Our preliminary analysis reveals that the use of Yoruba, Amharic, and Kinyarwanda is often sparse and not representative of natural, real-world conversation. On X (formerly Twitter), we observed infrequent conversational use of these languages. On Reddit, the situation is similar; for instance, the Yoruba subreddit primarily features requests for English-to-Yoruba translations or questions heavily code-switched with English, rather than authentic discourse [3]. This pattern of English dominance is also prevalent on subreddits for Amharic [4] and Kinyarwanda [5]. This low and unrepresentative digital presence creates a two-fold problem for both linguistic communities and technology development:

- **Scarcity of Authentic Training Data:** The lack of robust, conversational data in these languages online hinders the development and fine-tuning of effective language technologies. LLMs trained on existing web data are often ill-equipped to understand and generate these languages as they are used colloquially.
- **A Negative Feedback Loop:** The limited and skewed data available online leads to poor performance of AI models in tasks like language identification and translation. This poor performance can, in turn, discourage users from using their native languages on digital platforms, further exacerbating the data scarcity problem.

3 Related Work

The challenge of representing African languages in the digital age has been approached from several perspectives. One major area of focus has been the creation of datasets to improve NLP model performance. For instance, [2] developed a dataset of approximately one million human-translated sentences in eight African languages to fine-tune LLMs, demonstrating significant performance gains. Similarly, the Masakhane initiative organized NLP practitioners to build a named-entity recognition (NER) dataset for ten African languages, highlighting transfer learning potential and open research challenges [6]. Recognizing the foundational need for structured data, [7] developed a framework for systematic data collection and documentation for South African languages, which they used to build a language identification model. A parallel body of research examines the role of new media in language preservation and endangerment. Language is a complex support system essential for social function, and modern digital platforms offer new avenues for its documentation and revitalization [8]. Some scholars are optimistic; [9] suggests that platforms such as YouTube can enhance digital linguistic diversity by providing low-resource languages with broader reach through accessible video and music content. This aligns with the findings that social media platforms like blogs can be ideal spaces for cultural and intercultural learning, even if not specifically focused on African languages [10]. However, others express caution, arguing that the same forces of globalization and urbanization that threaten languages offline are amplified in the digital realm. [11] contends that the widespread use of dominant languages such as English and Mandarin contributes to the decline of indigenous languages, as younger generations often adopt languages that appear to offer greater economic and social opportunities. Building on this existing research, our study provides an empirically focused analysis of language use patterns on conversational platforms. We aim to quantify the digital presence of Yoruba, Kinyarwanda, and Amharic and evaluate how current AI technologies perform in identifying them. Our findings offer insights into the practical challenges and opportunities for improving the representation and processing of African languages in the digital age.

4 Methodology

The methodology followed in this study is described in Figure 1:

4.1 Data Sourcing

For this project, we extracted data from two different sources to understand how people interact and the language concentration on different platforms.

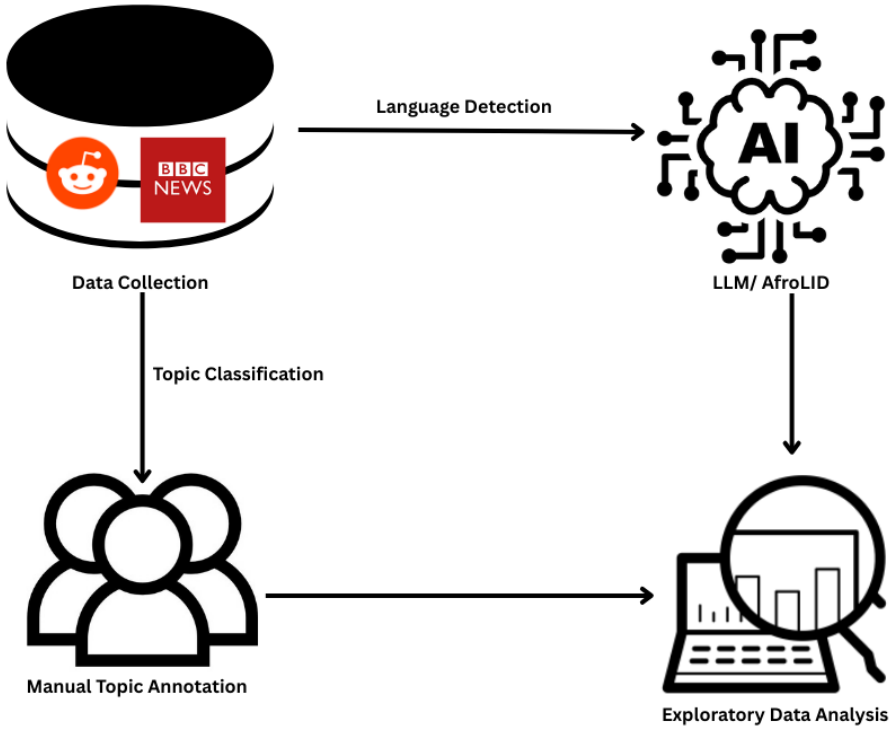


Figure 1: Workflow diagram of the study's methodology

4.1.1 Reddit

Due to the infrequent use of social media platforms like Reddit compared to Twitter or Facebook, we scraped three subreddits: r/yoruba[3], r/amharic[4], and r/Rwanda[5]. These subreddits correspond to the Yoruba, Amharic, and Kinyarwanda languages, respectively. We used PRAW[12] to scrape data for 365 days (one year) from each of these channels.

4.1.2 News Pages

Because English was the most dominant language in the scraped Reddit data, we sought news channels that post primarily in local languages. For Yoruba, we scraped BBC Yoruba[13]. For Kinyarwanda, we used Rwanda Broadcasting Agency[14], and for Amharic, we used Fana Broadcasting Corporation[15]. This approach provided a robust concentration of local languages. We also examined the social media pages of these news channels, where we found numerous comments in local languages on their Facebook posts.

4.2 Language Detection

In this research, we employed a two-pronged approach for language detection to effectively handle the linguistic diversity and potential code-switching present in our datasets. Firstly, we utilized AfroLID[16], a specialized language identification framework designed specifically for African languages, which covers 517 languages and varieties. This tool was crucial for accurately identifying the African languages in our corpus. Secondly, we observed that a significant portion of the posts were in English and AfroLID does not support English language detection, we initially considered incorporating LangDetect[17]. However, we found that LangDetect does not support the three African languages we are investigating. Given the need for a robust multilingual model that could handle both African languages and English in code-switched contexts, we instead opted to use the open-source multilingual large language model, Llama 3.3 70B. Although our target languages—Yoruba, Kinyarwanda, and Amharic—are not explicitly listed in its natively supported languages, the model's performance on common industry benchmarks and its design for multilingual dialogue made it a suitable choice for our exploratory analysis. This two-part strategy allowed for a comprehensive and robust detection process across our diverse datasets.

4.3 Topic Classification

To understand the context of what constitutes the most common news topics in these languages, we conducted a topic classification of the news articles. For this process, four native speakers—one for Yoruba, one for Amharic, and two for Kinyarwanda—sat down to annotate the titles by topic. The categories used for annotation included Business, Education, Sports, and Tourism, among others. This manual annotation process provided a robust understanding of the kinds of conversations and topics that are prominent in the news media for each language.

5 Results and Discussion

Our analysis reveals significant patterns in the digital representation of Yoruba, Kinyarwanda, and Amharic languages across different platforms and the varying performance of language detection tools on these datasets. The findings are categorized into three key areas: digital language usage on social media, a comparison of language detection models, and an analysis of prominent topics in news media.

5.1 Language Usage on Social Media

Our investigation into the usage of African languages on social media platforms revealed a pervasive trend of English dominance and code-switching. A qualitative analysis of the scraped Reddit data indicates that when these languages are used, they are often not in a conversational context. Instead, their usage is limited, primarily appearing in questions or for translation purposes, with a heavy prevalence of English. This observation aligns with the low total number of posts collected from the subreddits over a one-year period with Yoruba having 184 posts, Amharic having 165 posts and Kinyarwanda having 833 posts

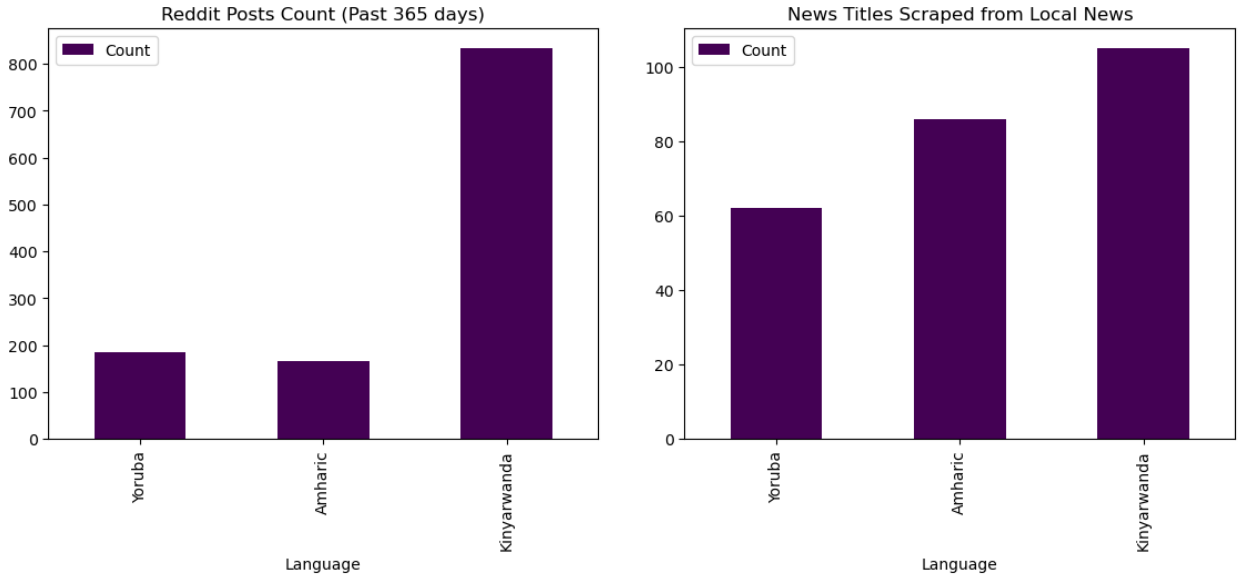


Figure 2: Comparison of Reddit Posts and News Articles Count. Left: Number of Reddit Posts. Right: Number of News Articles.

Given this sparsity and the code-switched nature of the Reddit data, we sought an alternative, more robust source of language data. We found a different dynamic on the social media pages of local news channels. When a post is initiated in a local African language on platforms like Facebook, it elicits a significant percentage of responses in the same local language as seen in Figure 3. This demonstrates that content generated in the local language, without code-switching, can stimulate authentic conversational engagement. In contrast, the news titles scraped from these sources yielded a higher concentration of clean, non-code-switched language data, with 62 posts in Yoruba, 86 in Amharic, and 105 in Kinyarwanda.

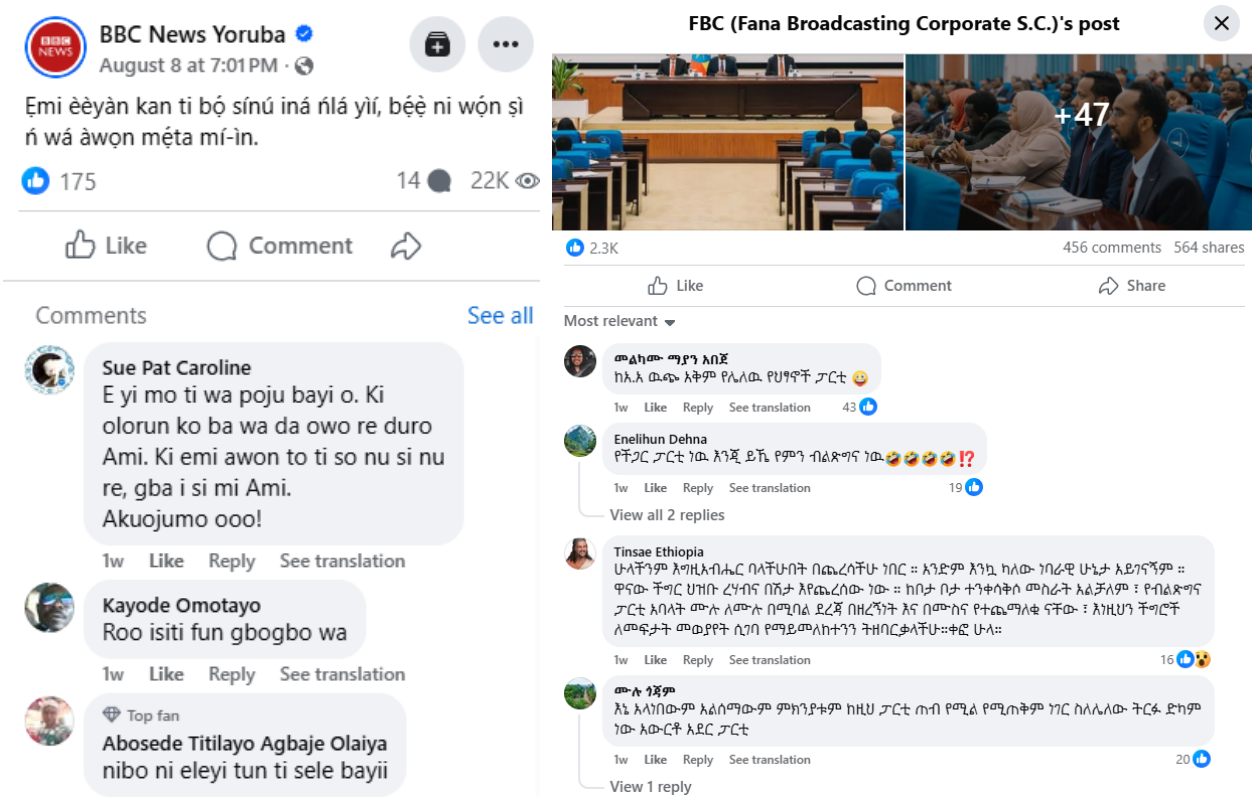


Figure 3: Comparison of BBC Yoruba and FANA Broadcasting Comments. Left: BBC Yoruba Comments. Right: FANA Broadcasting Comments.

5.2 Language Detection Performance

Our language detection analysis provides quantitative support for the qualitative observations we made about language usage. By comparing the results of AfroLID and a state-of-the-art multilingual LLM (Llama 3.3 70B), we gain a clearer picture of the data's composition and the models' respective strengths and weaknesses.

5.2.1 AfroLID Performance

As shown in Table 1, the AfroLID model identified a complex linguistic environment on Reddit. While it correctly identified the target languages in their respective subreddits, it also revealed a high degree of multilingualism and code-switching. For instance, the `r/Rwanda` subreddit, while primarily Kinyarwanda (`kin`, 66.43%), also contained a notable presence of other African languages like Swahili (`swh`) and Wolof (`wol`). This pattern was also evident in the `r/Amharic` and `r/Yoruba` subreddits, which showed significant contributions from other languages like Igbo (`ibo`), Somali (`som`), and Wolof (`wol`). In stark contrast, when applied to news media, AfroLID demonstrated near-perfect

Table 1: Afrolid - Reddit Language Detection

Subreddit	amh	ful	ibo	kin	lug	orm	som	swh	wes	wol	yor	zul
r/Amharic	27.42	0	4.03	38.71	0	0	8.06	16.48	0	11.29	0	0
r/Rwanda	0	0.29	3.73	66.43	0.14	0.14	4.88	15.06	0.14	8.61	0	0.57
r/Yoruba	0	0	15.65	9.57	0	0	9.57	6.96	0	14.78	43.48	0

accuracy. As detailed in Table 2, the model correctly identified 100% of the Yoruba and Amharic news posts and 97.14% of the Kinyarwanda news posts as their source languages. This highlights AfroLID's effectiveness on clean, non-code-switched text and confirms the value of news media as a source for training and evaluation data.

Table 2: Afrolid - NEWS Language Detection

Source Language	amh	flr	kin	run	wol	yor
Amharic	100	0	0	0	0	0
Kinyarwanda	0	0.95	97.14	0.95	0.95	0
Yoruba	0	0	0	0	0	100

5.2.2 LLM Performance

The LLM results, presented in Table 3, provide an important counterpoint. The LLM detected a significantly higher percentage of English content on Reddit compared to AfroLID, identifying 73.02% in `r/Amharic`, 97.45% in `r/Rwanda`, and 69.23% in `r/Yoruba`. While the LLM correctly identifies the primary non-English language in each subreddit (e.g., Yoruba at 28.21% in its respective subreddit), its high English classification rate suggests that it is categorizing much of the code-switched text and other languages as English, likely due to its broader, general-purpose training.

Table 3: LLM - Reddit Language Detection

Source	amh	ara	eng	kin	spa	swh	unk	yor
r/Amharic	25.4	0	73.02	0	0	0.79	0.79	0
r/Rwanda	0	0.28	97.45	2.26	0	0	0	0
r/Yoruba	0	0	69.23	0	2.56	0	0	28.21

Interestingly, the LLM performed exceptionally well on the news data, achieving 100% accuracy for all three languages, as shown in Table 4. This indicates that when presented with clean, monolingual text, the LLM is highly effective. However, its performance on the social media data underscores the challenge of using general-purpose models to navigate the nuanced, fluid linguistic environment of online African communities.

Table 4: LLM - News Language Detection

Source Language	Amharic	Kinyarwanda	Yoruba
Amharic	100	0	0
Kinyarwanda	0	100	0
Yoruba	0	0	100

5.3 News Topic Classification

To gain a deeper understanding of the content within our news datasets, we conducted a manual topic classification of news articles from each source. The results indicate that the news media content reflects distinct cultural and societal priorities as seen in Table 5.

The analysis reveals distinct content patterns across the three languages:

- **Amharic News**: The overwhelming dominance of business content at 40.7% suggests a strong focus on economic and commercial affairs in Ethiopian Amharic media. This is complemented by significant coverage of sports (16.28%) and politics (15.12%), indicating a well-rounded news ecosystem that balances economic reporting with civic and recreational interests. The complete absence of health and history coverage may reflect editorial priorities that favor immediate economic concerns over longer-term societal issues or historical context.
- **Kinyarwanda News**: While business content leads at 31.43%, the distribution is more balanced across categories compared to Amharic news. The substantial presence of entertainment (17.14%) and health coverage (12.38%) suggests a media landscape that addresses both informational and lifestyle needs of its audience. The equal representation of history and politics (both at 11.43%) indicates attention to both contemporary governance and cultural heritage, while the complete absence of sports coverage may reflect different recreational priorities or media resource allocation in the Rwandan context.
- **Yoruba News**: The near-equal emphasis on politics (27.42%) and diverse content categorized as "Others" (25.81%) reveals a media environment heavily focused on civic engagement and varied community interests.

Table 5: News Topic from Each Language (%)

Language	Business	Education	Entertainment	Health	History	Others	Politics	Sports
Amharic	40.7	1.16	8.14	0	0	18.6	15.12	16.28
Kinyarwanda	31.43	0	17.14	12.38	11.43	16.19	11.43	0
Yoruba	1.61	6.45	17.74	4.84	1.61	25.81	27.42	14.52

The strong presence of entertainment content (17.74%) and notable education coverage (6.45%) - which is absent in the other two languages - suggests a media ecosystem that serves both democratic participation and knowledge dissemination functions. The relatively balanced distribution across most categories indicates a more diversified news landscape compared to the business-heavy focus seen in the other two languages.

These distinct patterns likely reflect not only different cultural priorities but also varying economic conditions, political climates, and media industry structures within each linguistic community. The data suggests that Amharic media serves primarily economic information needs, Kinyarwanda media balances practical life concerns with civic awareness, and Yoruba media emphasizes democratic participation and educational content.

5.4 Discussion of Findings

The comparative analysis of AfroLID and the multilingual Large Language Model (LLM) sheds light on the complex dynamics of African languages in digital spaces. Our findings reveal two critical, interconnected insights. First, the online conversational platforms, even those clearly dedicated to specific African linguistic communities, are not reliable sources for clean, monolingual data. The high prevalence of English and the pervasive nature of code-switching are a direct reflection of the sociolinguistic realities of many urban African societies, where English often serves as a lingua franca for digital interaction. The data shows that even when members of these communities engage on platforms like Reddit, their interactions are heavily mediated by English. This creates a significant challenge for researchers, as the raw data does not fully represent the authentic, spoken language. It underscores a crucial need to rethink data collection strategies beyond simply scraping text from existing social media platforms. Second, the consistent high performance of both AfroLID and the LLM on news media sources demonstrates a clear path forward. These platforms offer a valuable, high-quality resource for data collection because their content is intentionally created in a cleaner, more formal, and monolingual style. This finding is particularly significant because it empowers African communities to be producers of high-quality digital content in their own languages. By leveraging these existing, professionally curated sources, we can build a robust foundation for language technologies that is more reflective of the languages as they are written and spoken in formal contexts.

5.5 Ethical Considerations

This research adheres to ethical guidelines for computational linguistics research using publicly available data collected from Reddit discussions and news media social pages in accordance with platform terms of service. To protect user privacy, we removed usernames from all collected posts, ensuring individual contributions cannot be directly attributed to specific users. Our analysis focuses on aggregated linguistic patterns, treating the data as representative samples of online language use while maintaining user anonymity. This methodology enables responsible advancement of African language technologies, contributing to discussions on improved natural language processing capabilities for African languages while respecting online user privacy.

5.6 Future Considerations

Based on these findings, we propose several avenues for future research that are specifically tailored to the unique linguistic landscape of Africa.

- **Model Development for Code-Switched Data**: Given that code-switching is a natural and integral part of digital communication for many Africans, future efforts must move beyond simply collecting clean data. The next frontier is to develop new models and fine-tune existing ones to better understand and process code-switched data. Instead of treating code-switching as noise, these models should be designed to recognize it as a legitimate and important linguistic phenomenon, thereby creating technologies that are more relevant and useful to the people who will use them.
- **Leveraging Audio Data**: Our findings highlight a disconnect between written social media content and the vibrant, multilingual reality of spoken language. A promising avenue for data collection is to look into

audio-based social media platforms. By employing strong Automatic Speech Recognition (ASR) technology tailored to African languages, we could transcribe spoken conversations. This approach would allow us to capture more authentic, conversational language use, including the nuances of intonation, rhythm, and natural code-switching that are lost in written text. This would provide a more representative dataset for building truly conversational LLMs.

- **Afro-Centric Benchmarking**: To measure true progress, we need to create benchmarks that are culturally and linguistically relevant. These benchmarks should not only test a model's ability to handle monolingual text but also its competence in managing tasks that involve code-switching, transliteration, and context-specific cultural references. By developing and using these benchmarks, we can ensure that language technologies are not just performant, but also deeply respectful and useful for the communities they are intended to serve.

6 Conclusion

This study has highlighted a critical divergence in the digital representation of African languages. While local news media provides a valuable source of clean, monolingual data for languages like Yoruba, Kinyarwanda, and Amharic, conversational platforms like Reddit are characterized by code-switching and data scarcity. Our findings demonstrate that current language detection tools, which perform well on formal text, struggle significantly with this informal, mixed-language content. This research underscores a fundamental challenge for AI development in low-resource language communities. We conclude that future research must move beyond the reliance on traditional, clean datasets and instead focus on developing models specifically designed to handle code-switched text, and audio conversations. This will be essential for creating inclusive and effective language technologies that truly reflect the dynamic and diverse linguistic realities of African users in the digital age.

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